

```
kubectl exec -it postgres-0 -- psql -U postgres -d mealdb
```

Flow:

```
Your Laptop
  ↓
kubectl
  ↓
Kubernetes Pod
  ↓
Run command inside container
  ↓
Open PostgreSQL shell
```

Now parts:

1. kubectl exec

Means:

```
Execute a command inside a running pod/container
```

Like SSH-ing into a server, but Kubernetes style.

2. -it

Actually two flags together:

-i

```
interactive
```

Keeps STDIN open so you can type.

-t

```
tty terminal
```

Gives you a proper terminal session.

Without `-it`, `psql` shell would not behave interactively.

3. postgres-0

This is the pod name.

From StatefulSet:

```
postgres-0  
postgres-1  
postgres-2
```

StatefulSets create stable pod names.

So Kubernetes now knows:

```
Run command inside pod postgres-0
```

4. --

Very important separator.

Means:

```
Everything after this is NOT kubectl command.  
Pass it to container.
```

Without -- , kubectl may think `psql` flags belong to kubectl.

5. psql

This is PostgreSQL CLI tool.

Equivalent of:

```
mysql  
mongo  
redis-cli
```

for Postgres.

It opens SQL shell.

6. -U postgres

Means:

Connect using PostgreSQL user postgres

This is DB username, not Linux username.

Equivalent SQL concept:

login as postgres user

7. -d mealdb

Means:

Connect to database named mealdb

Postgres server can contain many databases:

postgres
mealdb
testdb
analyticsdb

You are choosing one.

So full English sentence becomes:

Hey Kubernetes,
open an interactive terminal inside pod postgres-0,
then run PostgreSQL client,
login as user postgres,
and connect to database mealdb.